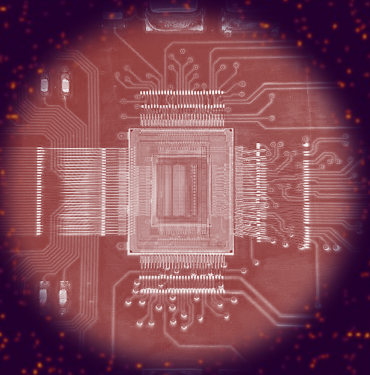
2026/10/20

Cryo-CMOS power management



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Introduction

- Llorenç Fanals-i-Batllori.
- BSc EE Universitat the Girona, Spain (9.52/10), MSc EE UPC Barcelona, Spain (9.69/10).
- Mixed-signal Design Verification Engineer at Apple, UK.
- Mixed-signal ASIC Design Engineer at SLAC/Stanford.



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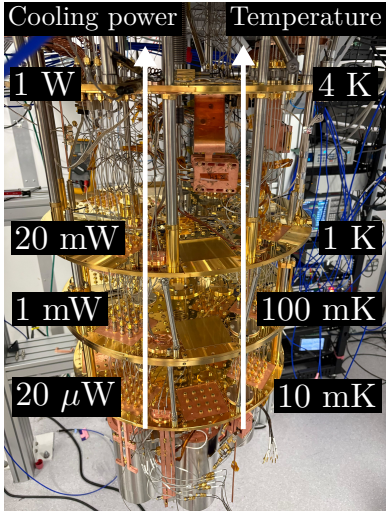
1. Cryogenic systems
2. MOS temperature behavior
3. Cryo LDO literature review
4. Design

1

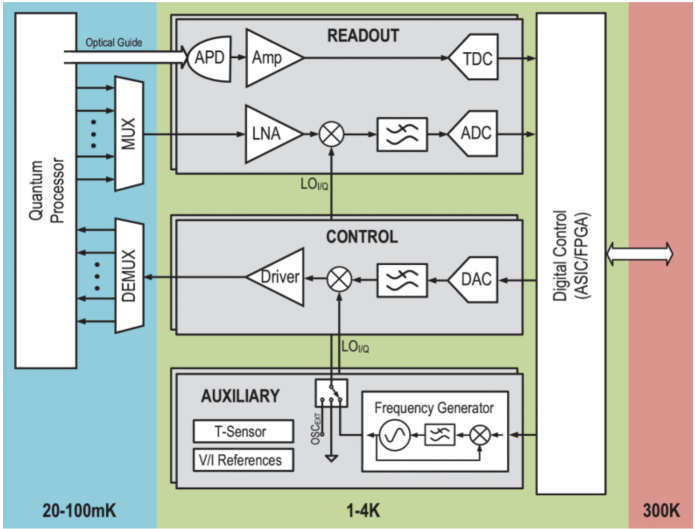
Cryogenic systems

Dilution refrigerator

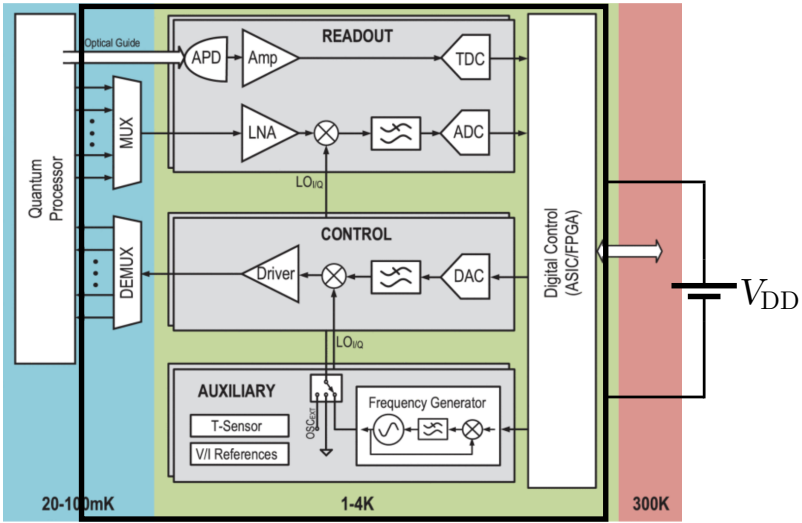
- The power budget scales approximately as $\propto T^2$.
- Useful quantum computers need thousands or millions of qubits.
- $k_B T \ll \omega\hbar$ for quantum effects to show up [5, Blais 2021].
- Cryo-CMOS at $\mathrm{m\,K}$ temperatures needs to be show near zero static power consumption.
- The $4\text{-}\mathrm{K}$ is regarded as a suitable stage for more power-hungry cryo-CMOS [9, Patra 2018].



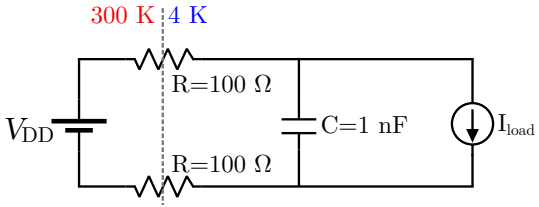
Cryo-CMOS motivation [9, Patra 2018]



The need for power [9, Patra 2018], adapted



Equivalent model, DC



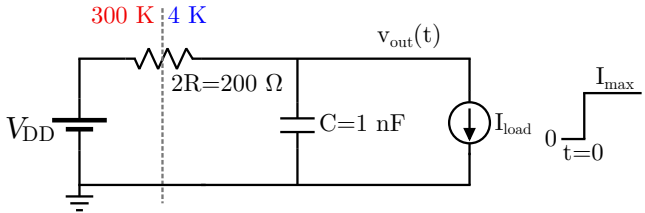
Efficiency:

$$\eta \triangleq \frac{P_{out}}{P_{in}} = \frac{V_{DD} I_{load} - I_{load}^2 \cdot 2R}{V_{DD} I_{load}} = 1 - \frac{2 I_{load} R}{V_{DD}} . \quad (1)$$

For $R = 100\,\Omega$, $V_{DD} = 1\,\mathrm{V}$, $I_{load} = 1\,\mathrm{mA}$, $\eta = 0.8$. The system stops working even for moderate loads!

- A single twisted pair is very resistive to minimize the heat load.
- Multiple twisted pairs in parallel are common, but this doesn't scale well.

Equivalent model, transient



$$\frac{V_{DD} - v_{out}(t)}{2R} = C\frac{\partial v_{out}(t)}{\partial t} + i_{load}(t)\,. \tag{2}$$

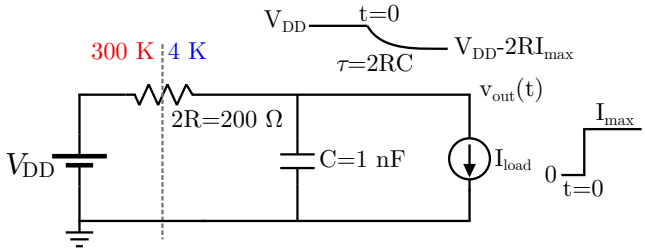
For a step on $i_{load}(t)$ from 0 to I_{max},

$$v_{out}(t) = V_{DD} - 2RI_{max}\left(1 - e^{-\frac{t}{2RC}}\right)\,. \tag{3}$$

For $R = 100\,\Omega$, $C = 1\,\mathrm{nF}$, $\tau = 100\,\mathrm{ns}$, slow!

Equivalent model, transient

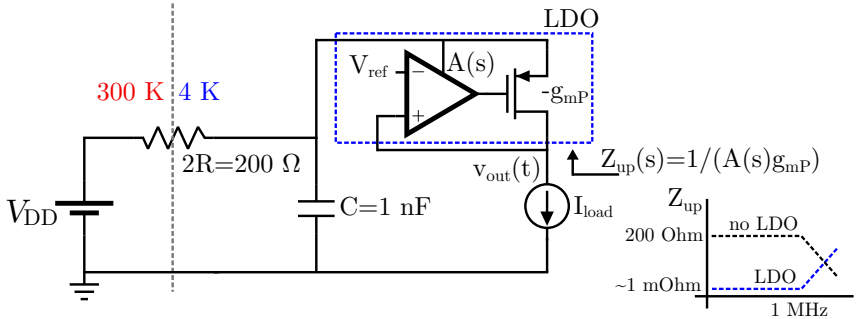
$$v_{\text{out}}(t) = V_{\text{DD}} - 2RI_{\text{max}}\left(1 - e^{-\frac{t}{2RC}}\right) . \tag{4}$$



If we increase the number of twisted pairs by N,

- $v_{\text{out}}(t \to \infty)$ gets closer to V_{DD}.
- τ is kept constant.

+ LDO



The output impedance of the LDO is very low, leading to

- fast transients.
- A well-regulated voltage, $\epsilon \triangleq V_{\text{ref}} - V_{\text{out}} = \frac{V_{\text{ref}}}{1+A(s)g_{mP}Z_{\text{load}}} = \frac{V_{\text{ref}}}{1+T(s)}$.

System-wise, the efficiency η is still poor. How to improve it?

+ buck converter [6, Fainé 2026]

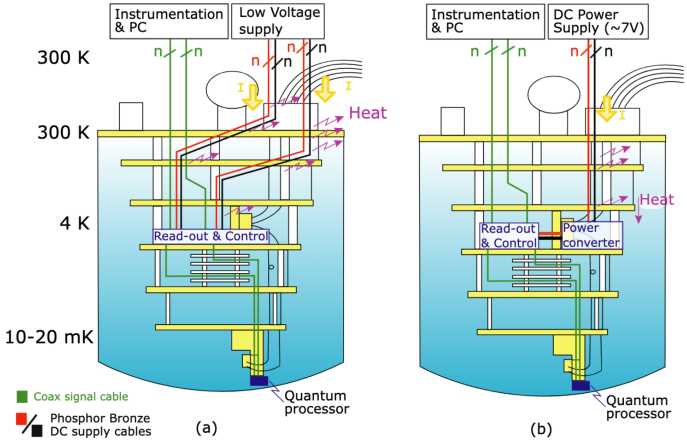
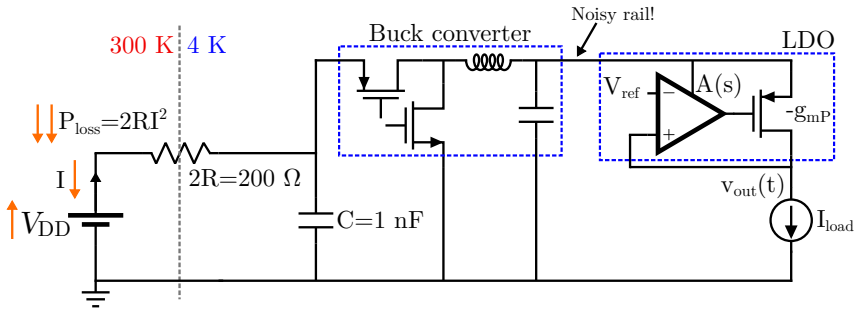


Fig. 1. QC power delivery architecture: (a) conventional architecture, (b) proposed architecture with local cryogenic voltage regulation.

+ buck converter

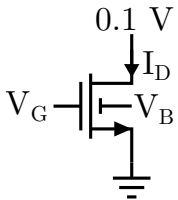


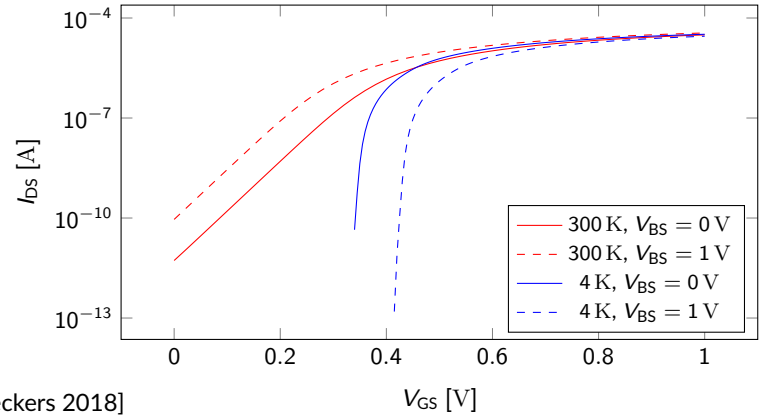
- V_{DD} is maximized to minimize the supply current. $P_{loss} = 2R \cdot I^2$ is minimized.
- An LDO is mandatory to clean the buck output ripple voltage.
- Ubiquitous architecture, found in almost all batery-operated devices.

2

MOS temperature behavior

MOS temperature behavior, 22-nm FDSOI





$300\,\mathrm{K} \rightarrow 4\,\mathrm{K}$ yields [3, Beckers 2018]

- An increase in threshold voltage $\Delta V_{TH} \approx 150\,\mathrm{mV}$.
- An increase in mobility $\frac{\mu_{4\,K}}{\mu_{300\,K}} \approx 2.5$.

MOS temperature behavior, 28-nm bulk [4, Beckers 2020]

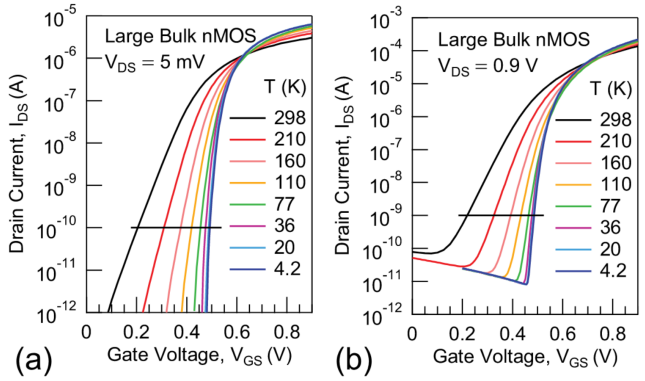
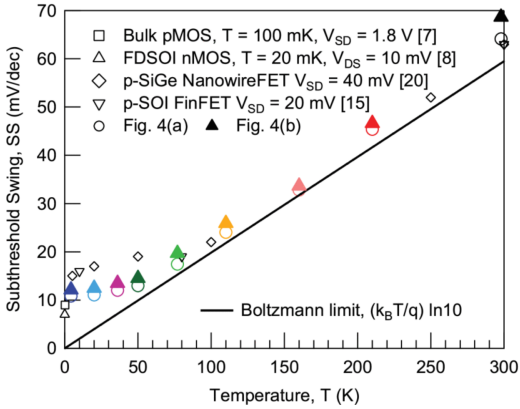


Fig. 4. Measured transfer characteristics down to 4.2 K in an *n*-channel, bulk-Si MOSFET with $W/L = 3\ \mu m/1\mu m$ from a commercial 28-nm bulk CMOS process, a) at low V_{DS} and b) high V_{DS}. Horizontal lines indicate current levels at which *SS* was extracted and shown in Fig. 1.

MOS temperature behavior, 28-nm bulk [4, Beckers 2020]



- As long as superconductivity is not achieved, MOS behaves the same below $35\,\mathrm{K}$.
- For calculations, take $T = 35\,\mathrm{K}$ as a saturation temperature.

MOS in weak inversion

In weak inversion,

$$I_{\mathrm{DS,wi}} = I_{\mathrm{D0}} \frac{W}{L} e^{V_{GS}/(nk_B T/q)} , \tag{5}$$

where

- $I_{DO} \propto e^{-V_{\mathrm{TH}}/(nk_B T/q)}$, take it as a constant.
- W, L: transistor effective width and length, respectively.
- $n(T)$: subthreshold slope factor, ~ 1.2 for FDSOI and 1.4 for bulk, increases at cryo.
- $k_B = 1.38 \times 10^{-23}\ \frac{\mathrm{m^2\,kg}}{\mathrm{s^2\,K}}$: Boltzmann's constant.
- T: temperature, in K.
- $q = 1.6 \times 10^{-19}\ \mathrm{C}$: charge of an electron.

MOS WI to SI transition [10, Sansen 2006]

In weak inversion,

$$I_{\mathrm{DS,wi}} = I_{\mathrm{D0}} \frac{W}{L} e^{V_{GS}/(nk_B T/q)} \ , \ \frac{g_{m,\mathrm{wi}}}{I_{\mathrm{DS,wi}}} = \frac{1}{nk_B T/q} = \frac{\ln(10)}{SS} \ . \tag{6}$$

In strong inversion,

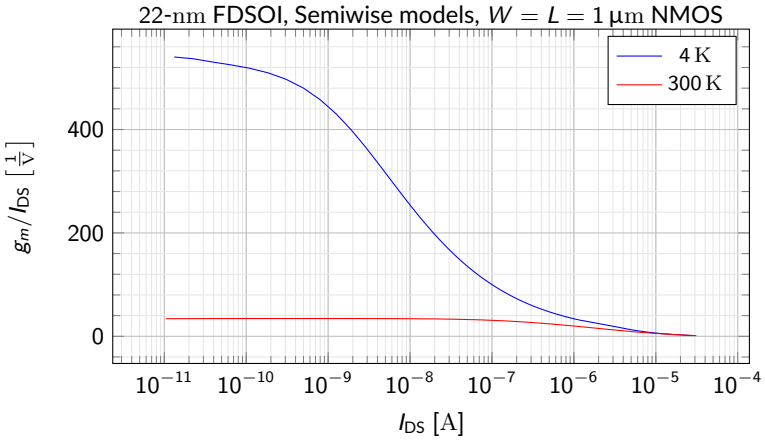
$$I_{\mathrm{DS,si}} \approx \mu_N C_{ox} \frac{W}{L} \left(V_{\mathrm{GS}} - V_{\mathrm{TH}}\right)^2 \left(1 + \lambda V_{\mathrm{DS}}\right) \ , \ \frac{g_{m,\mathrm{si}}}{I_{\mathrm{DS,si}}} = \frac{2}{V_{\mathrm{GS}} - V_{\mathrm{TH}}} \ . \tag{7}$$

The transition is at $\frac{g_{m,\mathrm{wi}}}{I_{\mathrm{DS,wi}}} = \frac{g_{m,\mathrm{si}}}{I_{\mathrm{DS,si}}}$:

$$\left(V_{\mathrm{OD}}\right)_t \triangleq \left(V_{\mathrm{GS}} - V_{\mathrm{TH}}\right)_t = 2n \frac{k_B T}{q} \ . \tag{8}$$

- $\left(V_{\mathrm{OD}}\right)_t$ = 72 mV at 300 K $\rightarrow$ $V_{\mathrm{OD}} = 200\,\mathrm{mV}$ is taken for strong inversion.
- $\left(V_{\mathrm{OD}}\right)_t$ = 8.4 mV at 4 K (assuming saturation at 35 K) $\rightarrow$ can achieve strong inversion with small V_{OD}, but be careful about V_{TH} mismatch!

g_m/I_{DS}



- In [4, Beckers 2020], $\min(SS) = 10\,\mathrm{mV/dec} \rightarrow g_m/I_{DS} = 230\,\frac{1}{V}$ and $g_m/I_{DS} \approx 500\,\frac{1}{V}$ at low I_{DS}.
- Large g_m/I_{DS} means a high current efficiency, but a slow speed.

Small-signal validity in weak inversion

$g_m = \frac{\partial I_{DS}}{\partial V_{GS}}$ is a linearization around the operating point.

How much error does this approximation yield?

$$I_{DS,wi} = I_{D0} \frac{W}{L} e^{V_{GS}/(nk_B T/q)} . \tag{9}$$

A Taylor expansion of $f(x)$ around x_0 is

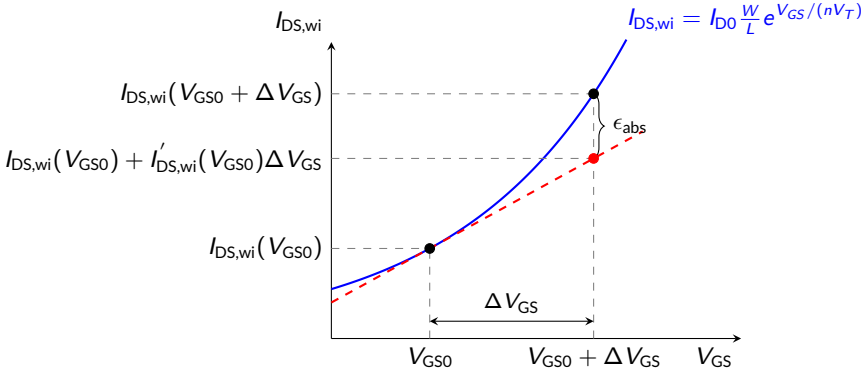
$$f(x) \approx f(x_0) + \frac{\partial f(x_0)}{\partial x^1} \frac{(x - x_0)^1}{1!} + \frac{\partial f(x_0)}{\partial x^2} \frac{(x - x_0)^2}{2!} + \frac{\partial f(x_0)}{\partial x^3} \frac{(x - x_0)^3}{3!} + \dots . \tag{10}$$

We can think in terms of absolute and relative errors:

$$\epsilon_{abs}(x) \triangleq \left(f(x_0) + \frac{\partial f(x)}{\partial x^1} \frac{(x - x_0)^1}{1!} \right) - f(x) . \tag{11}$$

$$\epsilon_{rel}(x) \triangleq \frac{|\epsilon_{abs}(x)|}{f(x_0) + \frac{\partial f(x)}{\partial x^1} \frac{(x-x_0)^1}{1!}} . \tag{12}$$

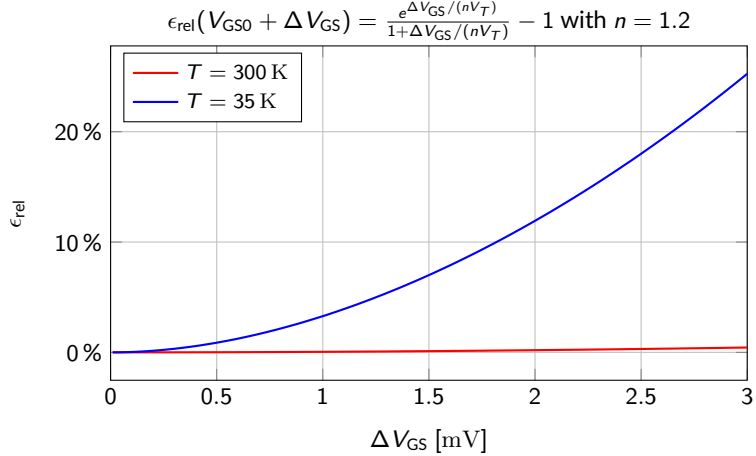
Small-signal validity in weak inversion



$$\epsilon_{abs}(V_{GS0}+\Delta V_{GS}) = I_{D0}\frac{W}{L}e^{V_{GS0}/(nV_T)}\left(1+\Delta V_{GS}/(nV_T) - e^{\Delta V_{GS}/(nV_T)}\right) . \tag{13}$$

$$\epsilon_{rel}(V_{GS0}+\Delta V_{GS}) = \frac{e^{\Delta V_{GS}/(nV_T)}}{1+\Delta V_{GS}/(nV_T)} - 1 . \tag{14}$$

Small-signal validity in weak inversion, ϵ_{rel}

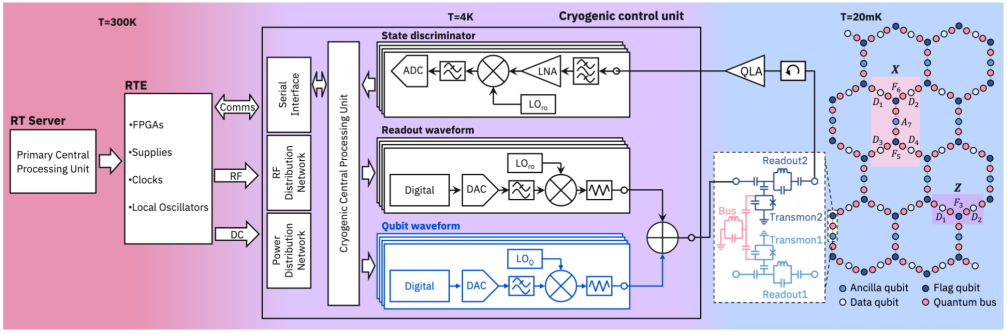


$\epsilon_{\text{rel}}(1nV_T) = 35.9\,\%$, $\epsilon_{\text{rel}}(2nV_T) = 146\,\%$. At cryo, small-signal approximations fail quickly!

3

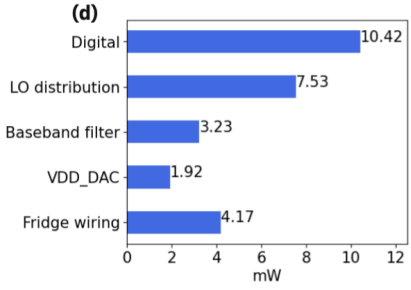
Cryo LDO literature review

IBM [12, Underwood 2023]



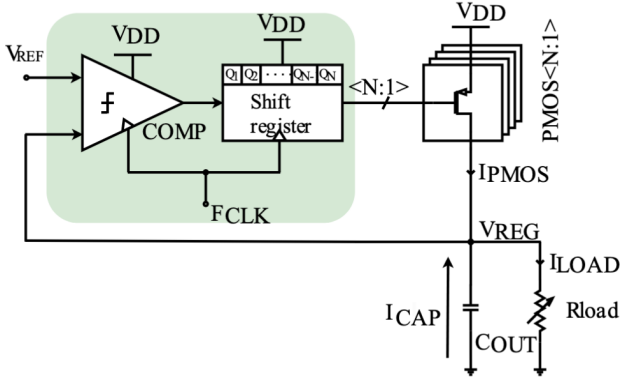
IBM [12, Underwood 2023]

ent experimental configurations: Run #1 was performed with the nominal device configuration and gate length, Run #2 was performed with a shorter gate length after adding additional low pass filtering to the supply voltages, and for Run #3 the on–chip attenuation was maxed



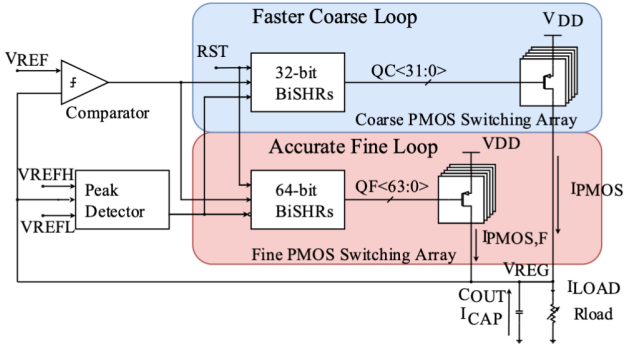
- A passive approach is valid too.
- Wiring losses amount to $4.17\,\mathrm{mW}$, amounting to $16\,\%$ of the total power consumption.

Jülich [2, Ashok 2025]



- Pros versus A-LDO: lower dropout, higher efficiency η, portable to advanced nodes, robust.
- Cons versus A-LDO: ripple in steady state, limited PSRR, speed (unless high power and ripple).

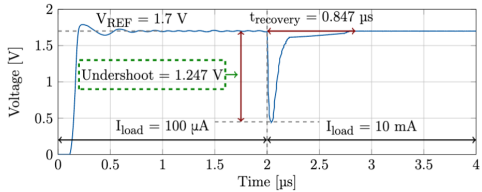
Jülich [2, Ashok 2025]



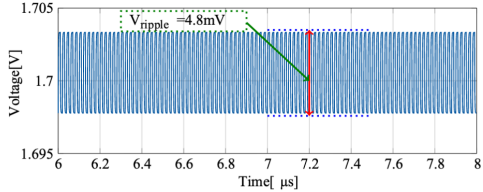
- Fine loop: enabled when $V_{\text{out}} \in [V_{\text{REFL}},\ V_{\text{REFH}}]$.
- Coarse loop: enabled when $V_{\text{out}} \notin [V_{\text{REFL}},\ V_{\text{REFH}}]$.
- $P_{\text{cons}} = 325\,\mu\text{W}$ with a clock frequency of $100\,\text{MHz}$, in 22-nm FDSOI.

Jülich [2, Ashok 2025]

- How to improve the transient response? Lower the depth of the flop chain (sacrifice accuracy) or run with a faster clock (increase power consumption).
- There is always some ripple in steady state. Acceptable for digital loads, not for high-precision analog or mixed-signal.

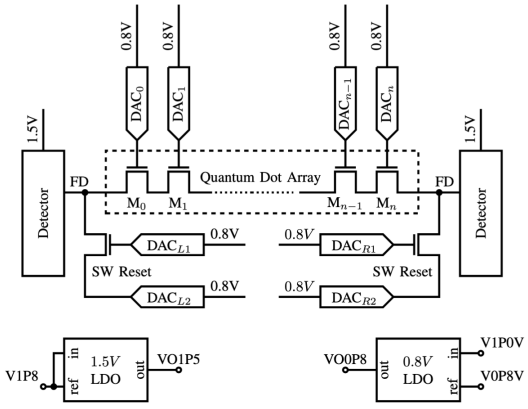


(b) Load transient response of the regulator for I_{load} of 100 uA to 10 mA



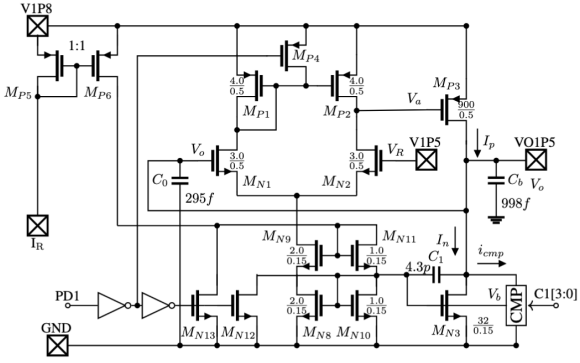
(c) Steady ripple or the load current of 10 mA

Equal1 [1, Andrade-Miceli 2021]



- A DAC will never be better than its analog reference. Quiet and regulated rails are needed.
- 1.5 V for thick oxide devices, mostly used for analog circuits.
- 0.8 V for thin oxide devices, mostly used for digital circuits, to benefit from $P_{\text{cons}} \propto V_{\text{DD}}^2$.

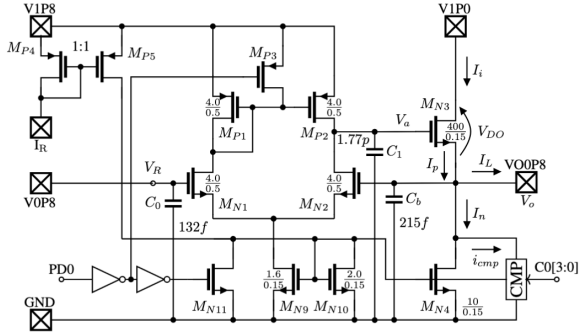
Equal1 [1, Andrade-Miceli 2021]



(b) 1.5-V LDO core.

- PMOS pass device.
- There is enough headroom to afford cascode devices.
- CMP is a tunable compensation current to facilitate stability.

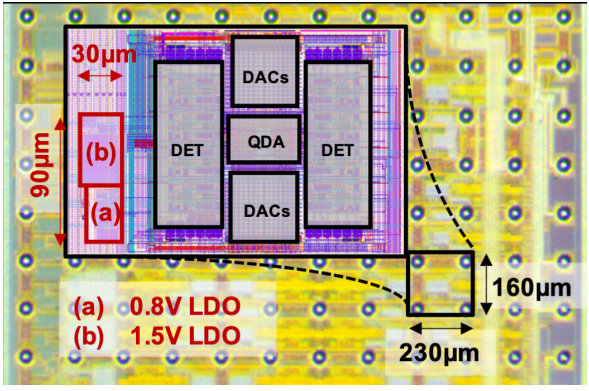
Equal1 [1, Andrade-Miceli 2021]



(a) 0.8-V LDO core.

- NMOS pass device.
- The error amplifier is operated via $1.8\,\mathrm{V}$, while the pass device sees $1.0\,\mathrm{V}$.
- Only thick-oxide devices are used, with a minimum length of $150\,\mathrm{nm}$.

Equal1 [1, Andrade-Miceli 2021]



- LDOs are sitting next to the DACs and circuitry.
- Capless means low area overhead.

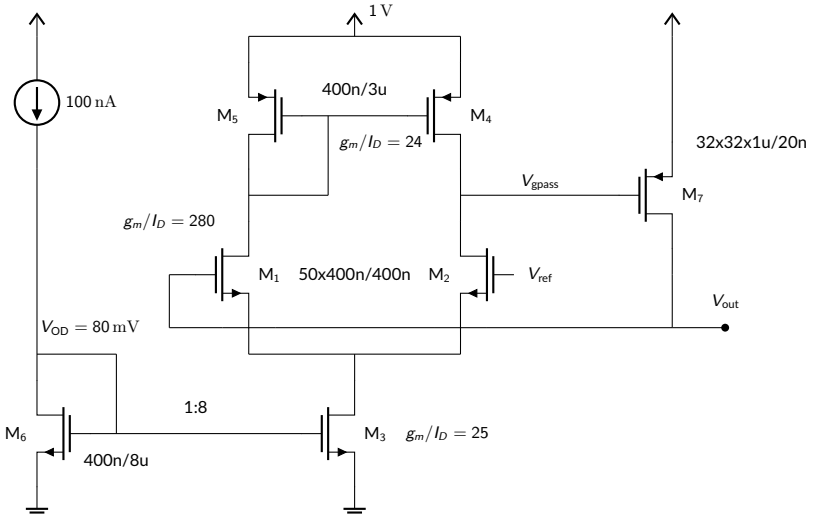
4

Design

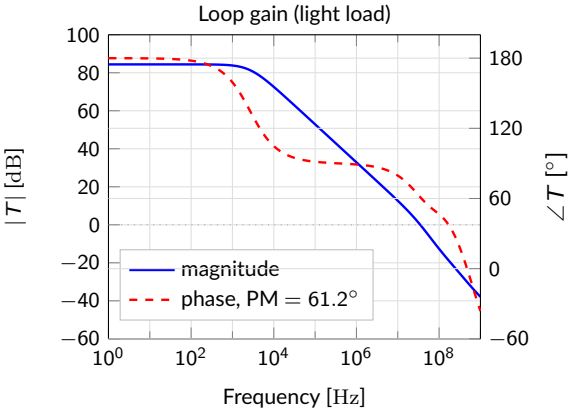
Design philosophy

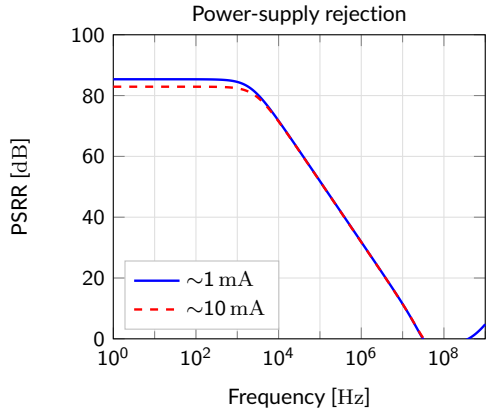
- Chose an analog LDO to have a ripple-free rail.
- Leverage weak inversion for a high g_m/I_D.
- Target $I_Q < 2\,\mu\mathrm{A}$ for compatibility with mK stages.
- Use dynamic large-signal circuits to aid transients.
- Accept the pass device dropout voltage as a tax to the power delivered to the load.

Error amplifier + pass device

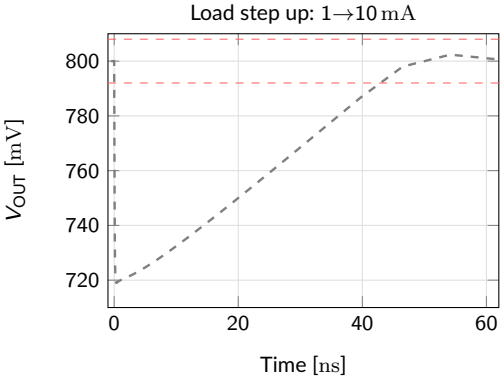


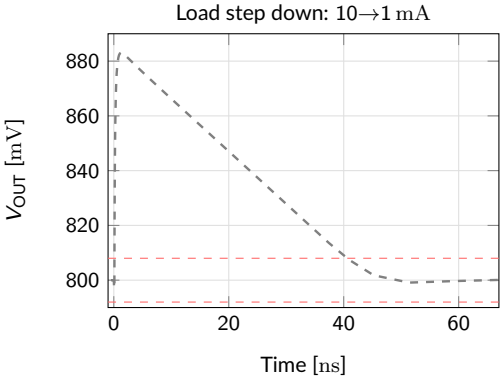
Frequency response



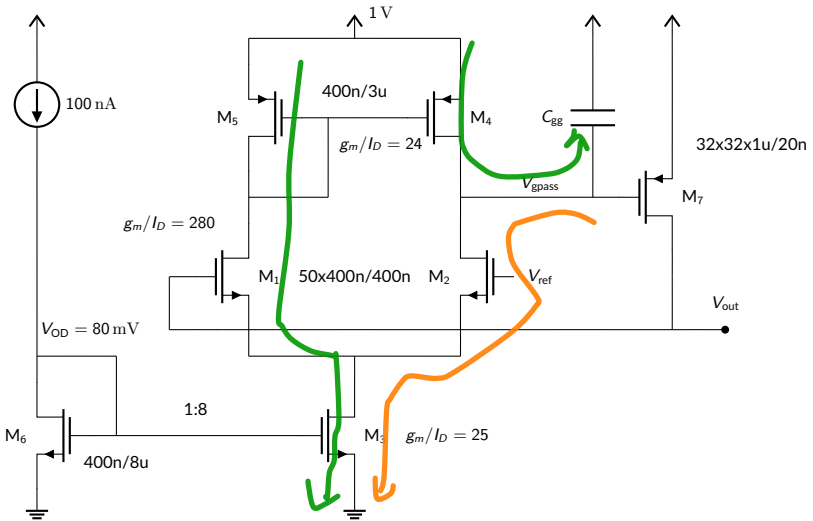


Transient response

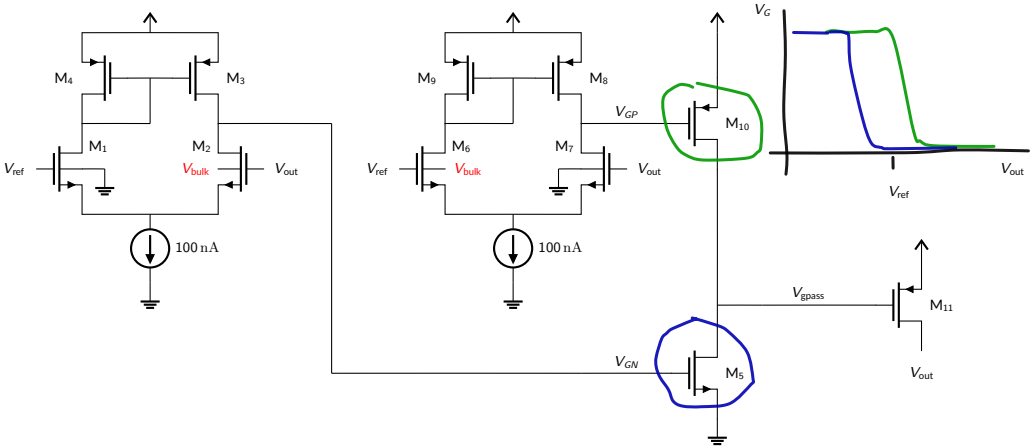




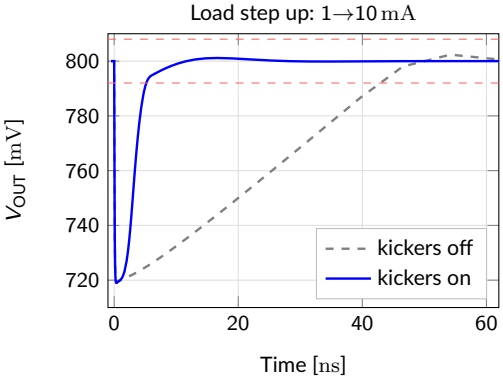
SR limitations

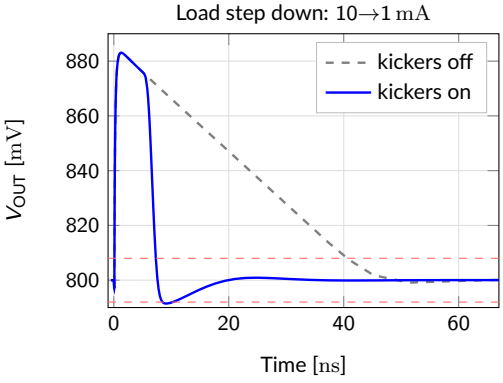


Transient improvement



Improved transient response





Comparison

	This work (sim.)	[1, Andrade-Miceli 2021]	[2, Ashok 2025]	[7, Huang 2020]	[8, Kundu 2019]	[11, Song 2021]
Technology	22 nm FDSOI	22 nm FDSOI	22 nm FDSOI	65 nm bulk	65 nm LP	40 nm bulk
Architecture	analog	analog	digital	mixed	mixed	mixed
Temperature	4 K	3.7 K	4 K	300 K	300 K	300 K
V_{IN}	1.0 V	1.0 / 1.8 V	0.9 V to 1.8 V	0.6 V	0.6 V to 1.2 V	0.6 V to 1.2 V
V_{OUT}	0.8 V	0.8 / 1.5 V	0.8 V to 1.7 V	–	0.4 V to 1.1 V	–
V_{DO}	200 mV	–	100 mV	50 mV	–	–
$I_{load,max}$	10.9 mA	64 mA	10 mA	10 mA	100 mA	–
I_Q	1.65 µA	∼10 µA†	325 µA	29 µA	∼800 µA‡	–
η_l	99.985 %	–	98.03 %	–	99.5 %	–
C_{out}	10 pF	**none**	–	–	40 pF	–
ΔI_{load}	1→10 mA	dc sweep	0.1→10 mA	0→10 mA	50 mA	104 mA
ΔV_{OUT}	81 mV	1.76 %	–	65 mV	108 mV	–
t_{rec}	10.5 ns	–	841 ns	–	1240 ns	6 ns
t_{settle}	10.5 ns	–	1150 ns	–	–	15 ns
PSR	–	–	–	−22 dB @ 1 MHz	–	–
PM	61°	–	–	–	–	–
Area	n/a	706 µm²	–	–	37 400 µm²	–
FoM§ (ps)	**0.014**	–	–	0.37	1.38	–

Table: †internal current reference; total I_Q not reported. ‡back-calculated from the reported FoM.

§FoM $= C_{out}\,\Delta V_{OUT}\,I_Q/\Delta I_{load}^2$; lower is better.

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